L31: Ranking and Selection

Simulation Without a Black Box

Outline

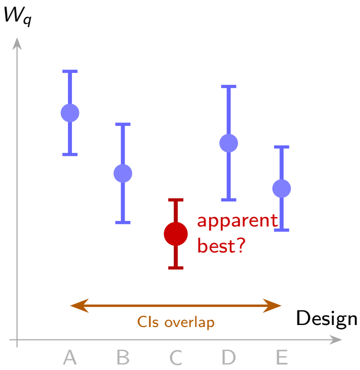
The Selection Problem: k Alternatives, Want the Best

Problem Statement

k designs with unknown means $\theta_1, \ldots, \theta_k$. Select the design with the *smallest* θ_i with probability $\geq P^*$ — the **PCS guarantee**.

Example: 5 staffing policies; find lowest W_q with $P^* = 0.95$.

- "Best observed" $\neq$ "statistically best"
- Lowest sample mean may be lowest *by luck*
- R&S procedures control the risk of choosing wrong



No guarantee without R&S

Picking $\min_i \bar{\theta}_i$ naively has no guaranteed PCS.

Why Pairwise t-Tests Inflate Error

FWER Inflation

Family-wise error rate (FWER): the probability of making *at least one* false declaration across all comparisons.

With $k = 5$ designs, there are $\binom{5}{2} = 10$ pairs. If each test uses $\alpha = 0.05$:

$$\mathrm{FWER} \approx 1 - (1 - 0.05)^{10} = 0.40$$

A 40% chance of at least one false declaration!

k designs	Pairs	FWER (nominal $\alpha = 0.05$)
3	3	0.14
5	10	0.40
10	45	0.90
20	190	≈ 1.00

Warning

Bonferroni correction (α/m per test) is overly conservative and wastes replications. Use dedicated R&S procedures instead — they are designed for the selection problem specifically.

The Right Guarantee

R&S procedures give:

$$P(\text{select true best}) \geq P^*$$

simultaneously across all k designs, without inflating the error rate.

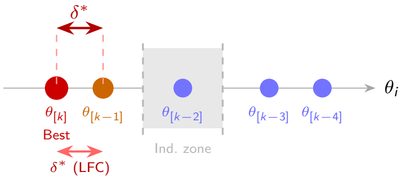
Indifference Zone Formulation

Indifference Zone Parameter δ^*

We don't care about differences $< \delta^*$. If the best design beats second-best by $\geq \delta^*$, we want to select it with probability $\geq P^*$.

Why δ^* matters:

- Without it, infinite samples needed to separate designs differing by 0.001 seconds
- Set by *domain knowledge*: "We don't care if W_q differs by less than 1 minute"
- Larger $\delta^* \Rightarrow$ fewer replications
- Smaller $\delta^* \Rightarrow$ more replications



Least Favorable Configuration (LFC)

$\theta_{[k-1]} - \theta_{[k]} = \delta^*$: second-best and best differ by exactly δ^*. R&S guarantees hold at the LFC.

Rinott's Two-Stage Procedure

Rinott Procedure

Stage 1 (pilot):

1. Run $n_0 \geq 2$ replications for each of the k designs
2. Compute sample variance s_i^2 for design i

Stage 2 (second stage):

1. Compute required total replications:

$$N_i = \max\left(n_0, \left\lceil \frac{h^2 s_i^2}{\delta^{*2}} \right\rceil\right)$$

2. Run $N_i - n_0$ additional reps for design i
3. Select design with best $\bar{\theta}_i$

h = Rinott constant (from table, depends on k, n_0, P^*).

```python
import numpy as np, scipy.stats as st

# Stage 1: pilot
n0 = 20
pilot = {i: run_design(i, n0)
         for i in range(k)}
s2 = {i: np.var(pilot[i], ddof=1)
      for i in range(k)}

# Rinott constant (from table)
# k=5, n0=20, P*=0.95 -> h ~ 2.8
h = 2.8
d_star = 1.0  # 1 minute IZ

# Stage 2: additional runs
N = {i: max(n0,
       int(np.ceil(h**2*s2[i]/d_star**2)))
     for i in range(k)}

all_means = {}
for i in range(k):
    extra = run_design(i, N[i]-n0)
    combined = np.concatenate(
        [pilot[i], extra])
    all_means[i] = combined.mean()

best = min(all_means, key=all_means.get)
```

Rinott Constant and Kim-Nelson Screening

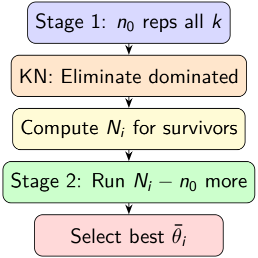
Rinott constant h at $P^* = 0.95$:

k	$n_0 = 10$	$n_0 = 20$	$n_0 = 40$
3	2.42	2.38	2.36
5	2.71	2.65	2.62
10	2.98	2.91	2.87
20	3.22	3.14	3.10

Larger $k \Rightarrow$ larger $h \Rightarrow$ more replications needed.

Kim-Nelson screening (KN): After the pilot stage, *eliminate* designs whose sample mean is provably worse than the current best by more than δ^*. Only advance surviving designs to Stage 2.

Benefit: Greatly reduces total simulation budget when many designs are clearly inferior.



Budget Trade-offs: δ^* and P^*

Trade-off table (M/M/1 clinic, $k = 5$, $n_0 = 20$):

δ^* (min)	P^*	Avg. N_i	Total budget
0.5	0.99	312	1560 reps
1.0	0.99	78	390 reps
1.0	0.95	52	260 reps
2.0	0.95	13	65 reps

Halving δ^* *quadruples* the required replications (because $N_i \propto 1/\delta^{*2}$). **Practical advice:**

1. Set δ^* from the decision context, *not* from the data ("What difference would change our recommendation?")
2. $P^* = 0.90$ or 0.95 is usually sufficient; $P^* = 0.99$ is expensive and rarely necessary
3. Use KN screening to eliminate obviously poor designs early

Common Mistake

Setting δ^* after seeing pilot results — this is circular and invalidates the PCS guarantee. Set δ^* before collecting any data.

Correct Process

1. Consult domain expert: "How small a difference in W_q would change your staffing decision?"
2. Set δ^* = that value.
3. Set P^* based on risk tolerance.
4. Look up h; run the two-stage procedure.

Key Takeaways

1. **Selection problem**: k designs, want best with $P(\text{correct}) \geq P^*$. Simple best-of-$k$ sample means has no PCS guarantee.
2. **Pairwise t-tests inflate FWER**: with $k = 10$, FWER $\approx 90\%$ at nominal $\alpha = 0.05$.
3. **Indifference zone δ^***: set from domain knowledge, not from data; small δ^* is expensive ($N \propto 1/\delta^{*2}$).
4. **Rinott procedure**: two-stage; Stage 1 estimates variance, Stage 2 runs $N_i = \lceil h^2 s_i^2 / \delta^{*2} \rceil$ reps.
5. **KN screening**: eliminate dominated designs after Stage 1 to reduce total simulation budget.
6. **Practical rule**: choose δ^* before running anything; $P^* = 0.95$ is usually sufficient; never set δ^* post-hoc.

End of Module M09 — Experimental Design

Next: Module M10 — Verification, Validation, and Advanced Topics.